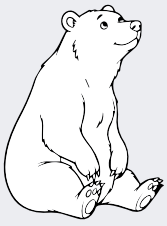


hibearnate
by Bear

Final Report
March 2026



Zain Ahmad ◦ Angus Bloch ◦ Francesco Rastelli ◦ George Ferguson

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Abstract

This report evaluates the viability of Hibearnate, Bear's first product range, an on/off blackout window film designed to eliminate light pollution and improve sleep quality - without the need for curtains or permanent installation.

Validation work mitigated three critical venture risks: technical feasibility, installation friction, and commercial viability.

- Laboratory fabrication confirmed that transparent-to-black film technology is technically achievable beyond supplier claims.
- Installation testing and iterative kit-of-parts refinement demonstrated that the system can be installed by users without professional assistance.
- In-house testing indicated measurable sleep improvements under blackout conditions, acceptable consumer price ranges above cost base, and early traction through digital marketing experiments.
- Paid acquisition testing suggested that customer acquisition cost could remain below projected gross profit per unit under reasonable conversion assumptions.

Following validation and testing, a **three-year financial projection** modelling production costs, pricing, marketing expenditure, and funding requirements was produced. The model indicates that the venture can sustain viable margins if conversion assumptions hold and manufacturing scale is achieved.

The project was delivered by a **four-person team** with responsibilities distributed across the report, financial modelling, website development, and video production, with all members contributed to experimentation, feedback, and final outputs.

The financial model is available at the linked **Excel file Here**

The website can be accessed at **lifebybear.com**

1 Private Section

1.1 Assumption Mapping

In the beginning of developing Hibearnate, we identified a range of assumptions across technical, behavioural, financial, and market dimensions to assess the viability of this venture:

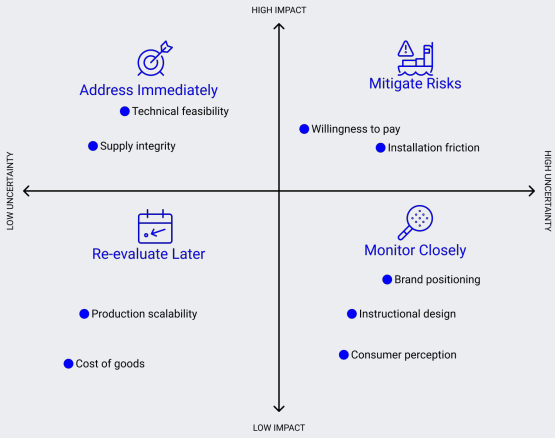
The key assumptions were namely:

1. Transparent-to-black **PDLC** film is **technically feasible** and producible.
2. Production of PDLC film can scale **beyond laboratory fabrication**.
3. Consumers perceive **light pollution** as a serious sleep-related problem.
4. Consumers are willing to **pay a premium** to solve this problem.
5. Installation can be completed by users **without professional assistance**.
6. Installation can be made **intuitive** through **instructional design**.
7. The cost of goods allows sustainable **gross margins**.
8. **Brand positioning** meaningfully influences desirability.
9. **Reliable supplier pathways** can be developed or controlled.

These assumptions were then evaluated in terms of their impact on venture survival and the level of uncertainty associated with each.

1.2 Risk Prioritisation

Each assumption was assessed using two criteria: the severity of its impact on the venture if incorrect, and the likelihood that it could fail. These were plotted on an impact-versus-uncertainty matrix.



Three risks emerged as the highest impact and highest uncertainty:

1. Technical feasibility and supply integrity
2. Installation friction
3. Commercial viability

If any one of these risks were not explored and mitigated, the venture would not be viable. So, validation efforts prioritise these three areas.

1.3 Risk 1: Technical Feasibility

Assumption Statement

Layman's terms: Could the desired product be manufactured?

If true transparent-to-black smart film cannot be obtained at scale, the core value proposition of Hibearnate collapses.

Underlying Assumptions

- Transparent to 99.9% black smart technology exists.
- Supplier claims are accurate and verifiable.
- A viable production pathway can be identified and costed.

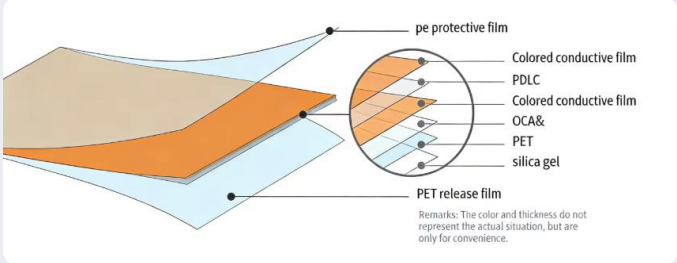
Experiment 1: Supplier Investigation

We conducted structured supplier research across Alibaba, AliExpress, and Taobao, with particular focus on Chinese-language supplier engagement to reduce pricing upcharges to English speaking customers. We identified major suppliers, compared quoted pricing, and ordered sample materials. Further analysis in Appendix C.

Continuing our supplier investigation we negotiated and purchased a test order of Wuhan Gani's film. The total cost of this was $151.94 (DDP to W6, UK) and allowed us to test 2 test sheets (10x16cm) and 2 larger form sheets (1mx1.5m). Unfortunately, the delivered sample was tinted rather than transparent-to-black, and the promised AC adapter was not supplied.

1.3 Risk 1: Technical Feasibility

It was clear that whilst the product became very dark, it was incapable of becoming transparent enough for everyday use. After questioning the supplier they explained (using the below image) that the ITO film they were using had a built in dark colouring (coloured conductive film) - therefore it would never be totally transparent.

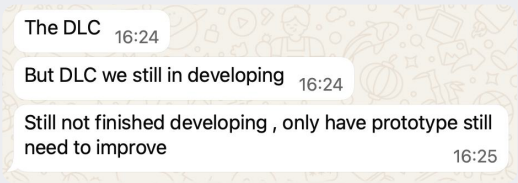


Realising that Wuhan Gani's product would not allow enough visible light transmittance (VLT) we once pivoted to look at alternative technology. At this stage we took the decision to move down 2 key routes for investigation of whether a suitable technology existed.

Route 1: Commercial route, look for the most advanced manufacturers. Here we discussed with Daryl Wiseman from Gentex their new SPD technology however we were unable to meet MOQ and current business model did not include providing film.



Following Gentex we connected with XXXX who is developing a new type of dimmable glass (neither SPD/EC/PDLC). However this technology was still in development and therefore not feasible for at this stage.



Following this experiment, we chose to explore a new lane: look for top researchers in the field of electrochromic film.

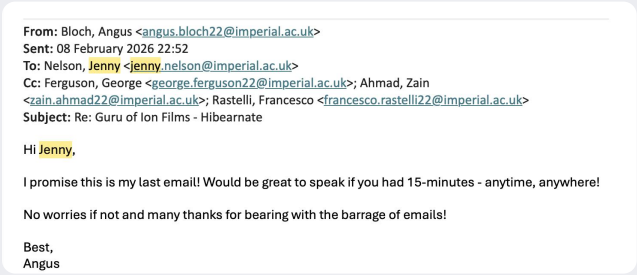
1.3 Risk 1: Technical Feasibility

Experiment 2: Academic Validation and Lab Fabrication

Here we escalated the validation process to a scientific level. We spoke to:

- Billy Wu - Dyson school of design engineering
- Sam Cooper - Dyson school of design engineering
- Jack Dawson - Kucernak Research Group
- Felix Watson - Electrical Science and Engineering Group, Imperial College London

The professors all recommended we talk to Jenny Nelson, a Professor in the Imperial College Physics Department, who's specialises in EC film. This process required persistence as Jenny is extremely busy, but following multiple emails and two in-person visits we successfully managed to meet Jenny and her team, learning key manufacturing insights for EC films. See Appendix X for full Jenny Nelson correspondence.



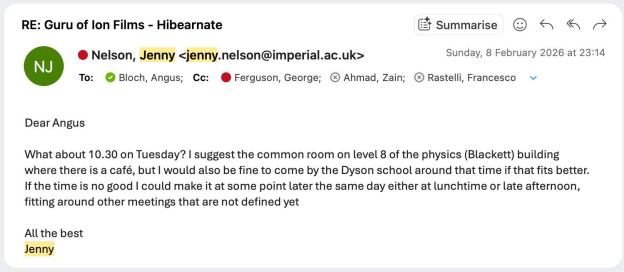
From: Bloch, Angus <angus.bloch22@imperial.ac.uk>
Sent: 08 February 2026 22:52
To: Nelson, Jenny <jenny.nelson@imperial.ac.uk>
Cc: Ferguson, George <george.ferguson22@imperial.ac.uk>; Ahmad, Zain <zain.ahmad22@imperial.ac.uk>; Rastelli, Francesco <francesco.rastelli22@imperial.ac.uk>
Subject: Re: Guru of Ion Films - Hibearnate

Hi Jenny,

I promise this is my last email! Would be great to speak if you had 15-minutes - anytime, anywhere!

No worries if not and many thanks for bearing with the barrage of emails!

Best,
Angus



RE: Guru of Ion Films - Hibearnate

Summarise

NJ

Nelson, Jenny <jenny.nelson@imperial.ac.uk>

Sunday, 8 February 2026 at 23:14

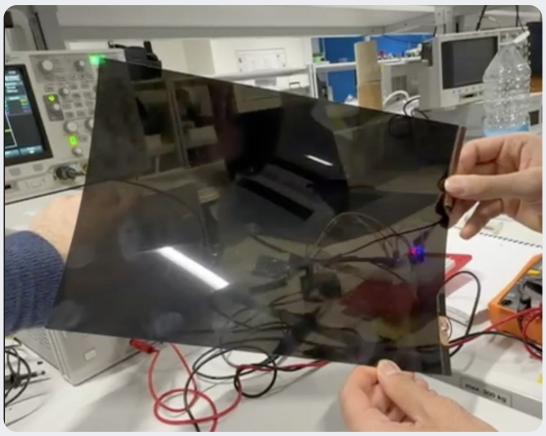
To: Bloch, Angus; **Cc:** Ferguson, George; Ahmad, Zain; Rastelli, Francesco

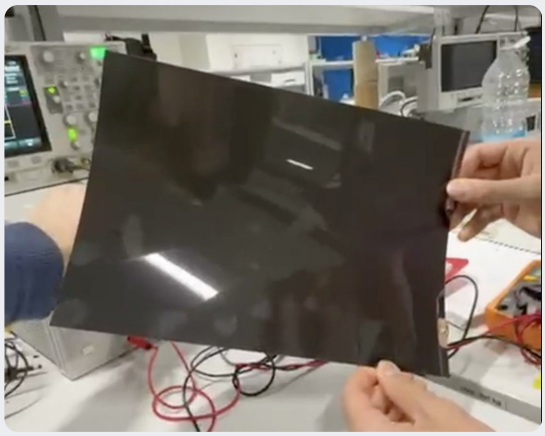
Dear Angus

What about 10.30 on Tuesday? I suggest the common room on level 8 of the physics (Blackett) building where there is a café, but I would also be fine to come by the Dyson school around that time if that fits better. If the time is no good I could make it at some point later the same day either at lunchtime or late afternoon, fitting around other meetings that are not defined yet

All the best
Jenny

Through laboratory access and collaboration, a working PDLC sample was fabricated and successfully demonstrated switching behaviour.





The fabrication process clarified required materials, equipment, and production considerations. We documented the equipment required for production-grade fabrication and integrated these inputs into our financial modelling (see Profit and Loss excel file "Manufacturing Facility" tab).

Learning

- We confirmed that transparent-to-black EC technology is technically feasible.
- Determined the supplier market is opaque and that relying on unverified commercial suppliers presents unacceptable risk.
- Direct technical oversight, potentially creating the first UK manufacturing plant of EC film has potential as it would be the first in the UK.

Remaining Uncertainty

- Yield rates at small-batch scale have not yet been tested, including long-term durability and environmental stress resistance.
- Establishing local production would reduce supplier dependency but require significant upfront capital expenditure.

The next stage involve addressing these uncertainties.

1.4 Risk 2: Installation Friction

Risk Statement

If installation is perceived as complex, risky, or technically intimidating, particularly among renters, adoption will fail regardless of technical performance.

Underlying Assumptions

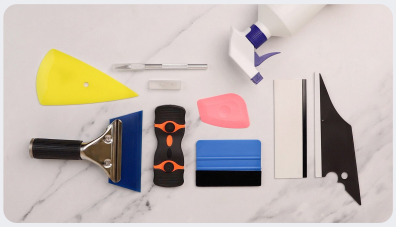
- Users prefer DIY installation over professional installation.
- Instructional clarity can compensate for technical complexity.
- Trimming and alignment for our film can be made intuitive.
- Standard kit of parts can be defined that is suitable for any type of customer.

Experiment 1: Competitive Teardown and Benchmarking

We purchased multiple sheet-based DIY products, including vinyl wraps and screen protectors, to understand how successful consumer brands reduce installation friction.

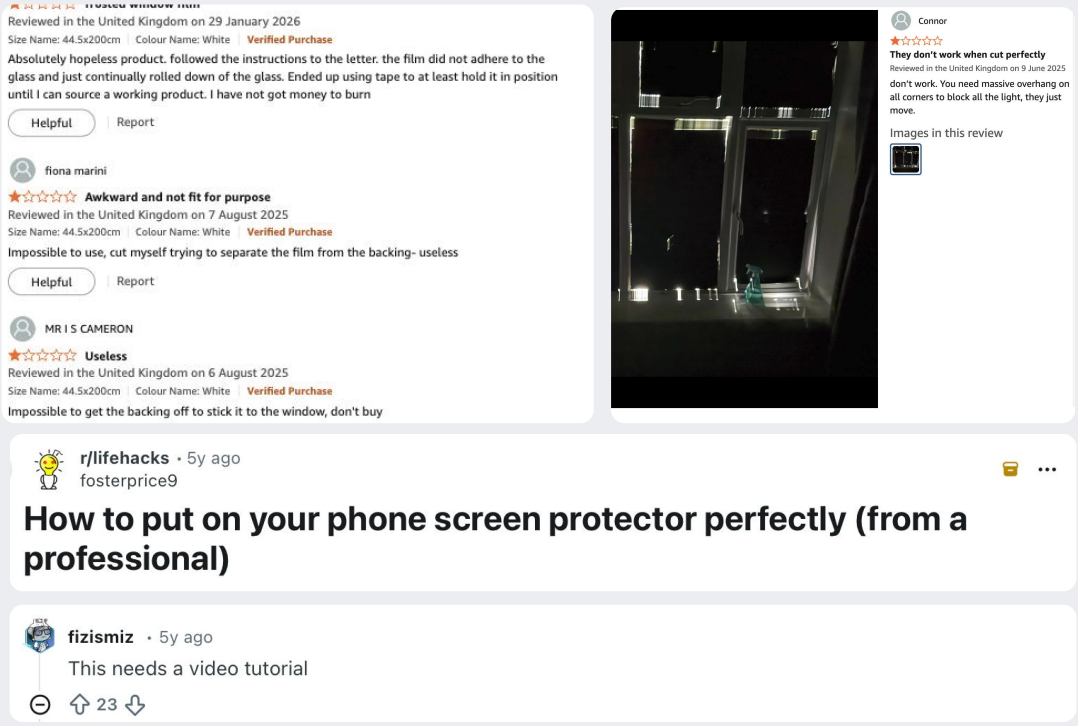






1.4 Risk 2: Installation Friction

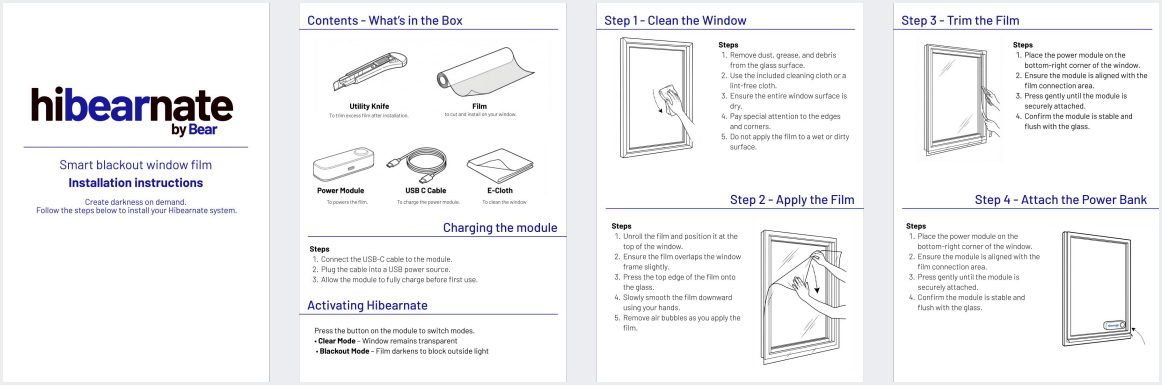
We studied their kit-of-parts, annotated their instruction manuals, and reviewed customer feedback online to identify common pain points.



We found that visual-first instructions consistently outperformed text-heavy manuals, alignment guides reduced anxiety, and trimming stages generated the most user complaints.

Experiment 2: Internal Installation Simulation

We developed Version 1 of the Hibearnate instruction manual and assembled a proposed kit-of-parts



We conducted an internal installation simulation with 12 participants using a small section of PDLC film applied to glass surfaces. Participants were asked to follow the instructions independently while we observed friction points.

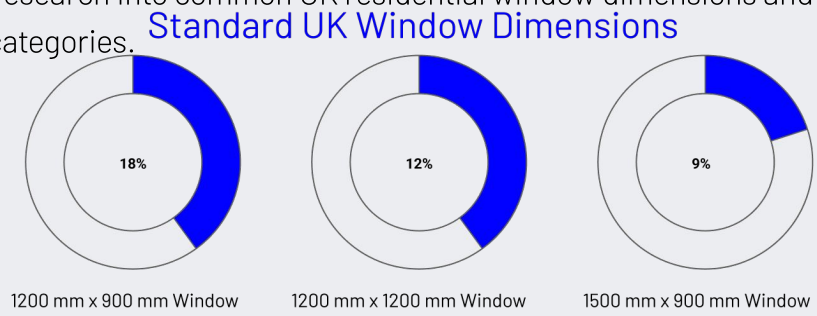
1.4 Risk 2: Installation Friction

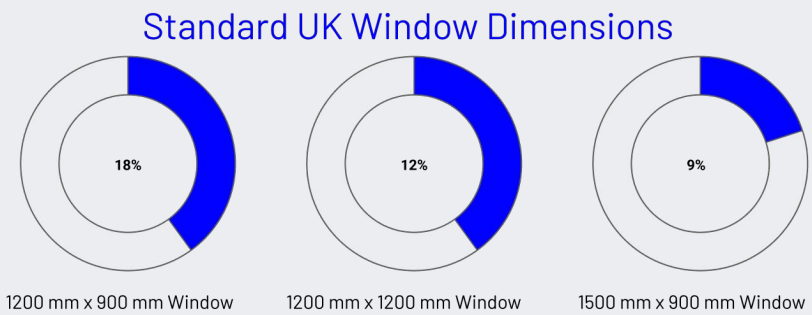
Observations

- The supplied wiping tool was insufficient across different window conditions. Dust and residue were not consistently removed, affecting adhesion.
- Not all participants were comfortable using a trimming blade. In some cases, this resulted in hesitation or minor surface scratching.
- Alignment was inconsistent when window dimensions were not clearly defined.
- Participants were unsure whether their window dimensions were suitable for the sheet provided.

Learnings

1. The kit-of-parts was updated to include both a dry wiping tool and a wet cleaning wipe to ensure proper surface preparation.
2. The trimming tool was redesigned to be more blunt and safer, reducing risk of scratching and increasing user confidence.
3. We conducted research into common UK residential window dimensions and identified three dominant size categories.





We found that most windows in the uk are not standard sizes but that these 3 are the most common. Based on this, we defined three standard Hibearnate sheet sizes (Small, Medium, Large) aligned with common UK window dimensions with these films having a coverage of 85% of windows after trimming. This reduced ambiguity around trimming and alignment.
Installation friction was therefore reduced not only through instructional clarity but through physical product redesign and SKU standardisation. This was also reflected in the P&L.

1.4 Risk 2: Installation Friction

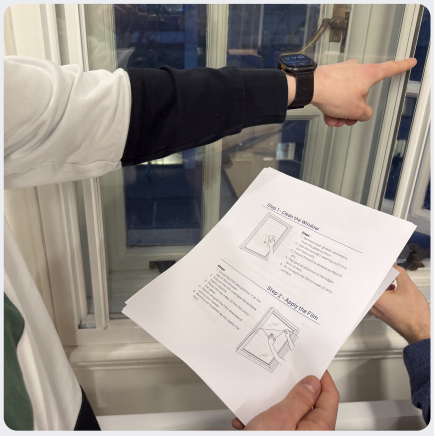
Experiment 3: A/B Instruction Manual Testing

Following the product and sizing refinements, we developed two revised instruction manual versions (Version A and Version B). Each participant received identical materials but only one manual version. Installation time, observed confusion points, and verbal feedback were recorded.

Participant Comparison

Characteristic	Installation Time (s)	Confusion: Alignment	Confusion: Trimming	Questions Asked
Version A	575-640	Yes	Yes/No	2-4
Version B	430-470	No	No	0-1





Version B reduced average installation time by **25%** and significantly reduced reported confusion compared to Version A.

Participants reported clearer sequencing, improved confidence during alignment, and less hesitation at the trimming stage.

Learning

The biggest lessons were:

- We found that installation friction was driven not only by instructional clarity but also by tool design and dimensional ambiguity.
- Improving visual hierarchy, simplifying trimming tools, and pre-defining sheet sizes significantly reduced hesitation and increased completion consistency.

Hence, through this iterative testing, we:

- Refined the kit-of-parts
- Standardised SKU sizing based on common UK window dimensions
- Optimised the instruction manual.

1.5 Risk 3: Commercial Viability

Risk Statement

If Hibearnate does not create sufficient perceived value to justify a premium price, or if customers cannot be acquired at a sustainable cost relative to margin, the business model fails.

This risk combines three elements:

- Does it create real value?
- Will people pay?
- Can we acquire them efficiently?

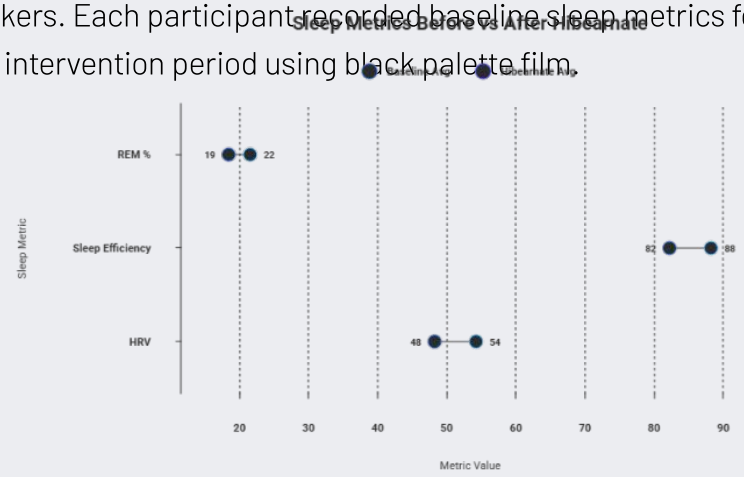
Underlying Assumptions

- Blackout conditions produced by Hibearnate meaningfully improve sleep quality in real residential environments.
- Consumers perceive improved sleep quality as a sufficiently valuable outcome to justify paying a premium price.
- The acceptable market price for Hibearnate exceeds the estimated cost of goods, allowing sustainable gross margins.
- Digital channels such as short-form social media and targeted advertising can generate early-stage attention and customer interest.
- Customers can be acquired at a cost lower than projected gross profit per unit.

Experiment 1: Functional Sleep Impact (Value Validation)

While substantial academic literature demonstrates that nocturnal light exposure disrupts circadian rhythm and REM sleep, our objective was not to re-prove the science of light hygiene. Instead, we aimed to validate whether Hibearnate's implementation of blackout in real residential settings produced measurable user-level benefit.

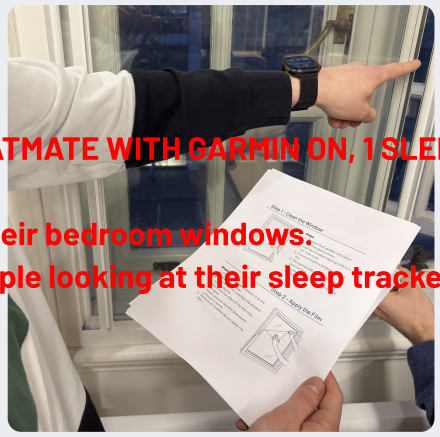
We conducted a 14-day pilot sleep test with three participants using a mixture of WHOOP/ Garmin/Aura Ring trackers. Each participant recorded baseline sleep metrics for week 1, followed by a blackout intervention period using black palette film.



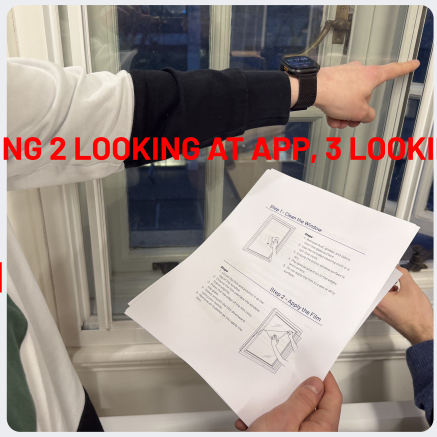
1.5 Risk 3: Commercial Viability



[ANGUS: TAKE PHOTO OF FLATMATE WITH GARMIN ON, 1 SLEEPING 2 LOOKING AT APP, 3 LOOKING AT WATCH]



Hibearnate film installed on their bedroom windows.



[insert Photo: 3 different people looking at their sleep trackers]

Metrics recorded: REM %, Sleep efficiency, HRV.

Participants experienced an average REM increase of 3% and sleep efficiency improvement of **6%**. All reported subjective improvement in sleep quality.

This experiment served three purposes:

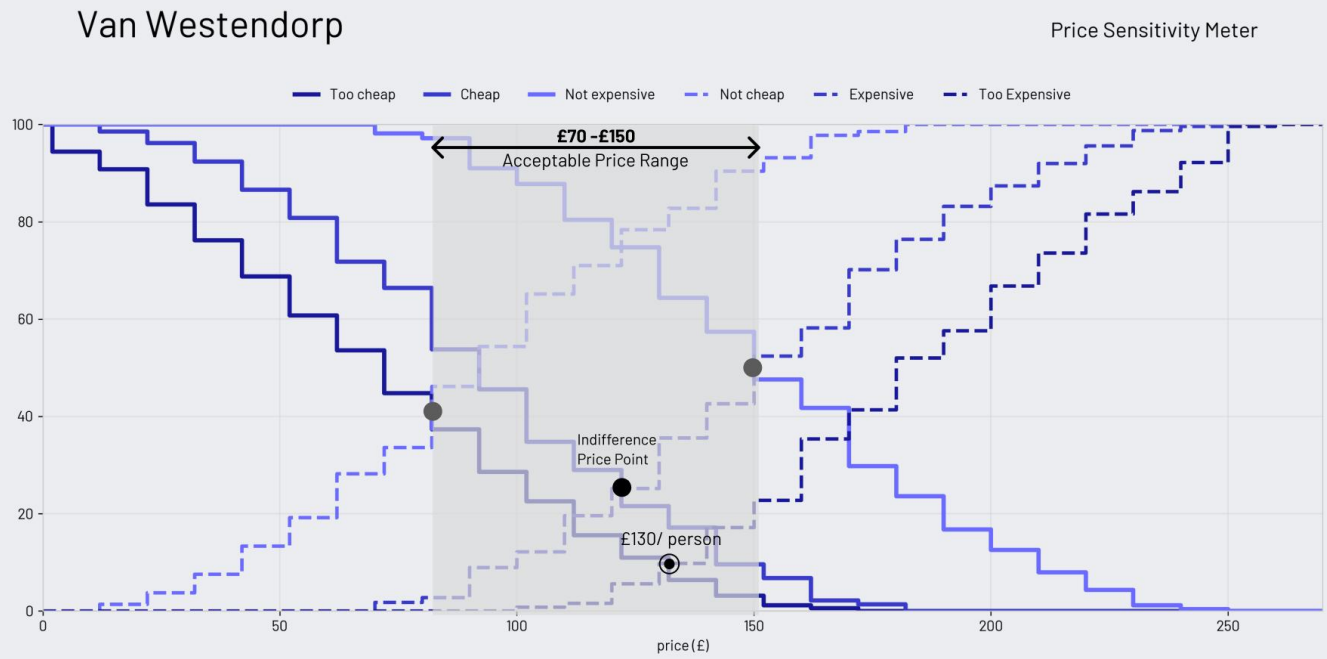
1. To confirm that our specific blackout implementation meaningfully reduces ambient light leakage in typical UK residential environments.
2. To generate venture-specific quantitative evidence supporting the value proposition.
3. To strengthen pricing justification by linking product performance to measurable outcomes.

Limitations include small sample size, lack of placebo control, and participant awareness of intervention. A larger, partially blinded study is planned.

1.5 Risk 3: Commercial Viability

Experiment 2: Van Westendorp Price Sensitivity Analysis

Here, we tested the assumption that consumers will accept a premium price point that generates sustainable margin above COGS to allow viable gross margin. To do so, we conducted a Van Westendorp price sensitivity survey use Prolific (see term 1 deck) in which participants were asked to identify price thresholds for "too cheap," "cheap," "not expensive," "not cheap," "expensive," and "too expensive."



The intersection point indicated an optimal price of **£130**. Our estimated COGS, derived from supplier analysis and lab-based fabrication inputs, is £23.18/£40.90/£52.59 (S/M/L). Assuming that £130 would be optimal for Large window (pessemistic estimation), and at an AoV of 2x SKU RRP (as on avg. 2 windows /room) we generate a projected CM1 of 59.54%. See snippet from COGS slide P&L.

6 **Calculate Total Cost**

This section combines the total cost for each Hibearnate SKU.

The total cost is computed by the standardised cost of each *auxilary* and *key* material, multiplied by The Material Quantities calculated in seciton 3.

Small		Medium		Large	
£	23.18	£	40.90	£	52.59

This indicates that perceived acceptable pricing exceeds cost base sufficiently to support marketing and scaling costs. However, this reflects stated willingness, not observed purchasing behaviour. Real conversion validation is required. Next step: deposit-based pre-order test.

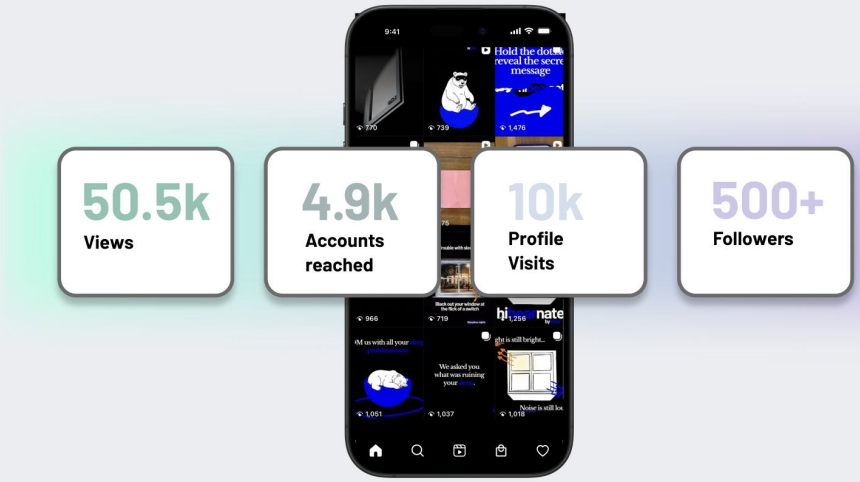
1.5 Risk 3: Commercial Viability

Experiment 3: Organic Brand Exploration

As a marketing-leaning team, we wanted to test whether Hibearnate could generate early-stage attention through organic short-form content.

We produced 15 TikTok and Instagram Reels exploring different storytelling approaches, tones, and visual formats. These were not structured A/B tests but exploratory content aimed at understanding baseline engagement and brand viability.

Social media traction in the last 30 days



This experiment allowed us to assess whether the concept resonated at a top-of-funnel level and whether we could generate attention without paid amplification. We observed consistent engagement and steady follower growth, indicating early-stage audience interest. While organic engagement does not equate to demand, this validated that the problem narrative had traction potential and justified further structured testing.

Experiment C2: Creative A/B Messaging Test

Following exploratory organic posting, we conducted structured A/B testing to determine which positioning angle resonated most strongly.

We tested three distinct framings:

- *"Struggling to block out the light?" (problem-led)*
- *"Smart Blackout Window" (feature-led)*
- *"Ditch the Eye Masks" (comparison-led)*

For each, we created subtly different ad creatives and tracked engagement metrics including:

- *Click-through rate*
- *Engagement rate*
- *Saves and comments*



1.5 Risk 3: Commercial Viability

Comparison-led (ditch the eye masks) messaging consistently generated higher click-through and deeper engagement compared to feature-led framing. This provided a clear learning: emotional, problem-centric positioning outperformed technical framing. Ditch the eye-masks was thereby used as a tag line throughout our marketing pieces.

Experiment D — Paid Acquisition Test (Cost per Waitlist Signup)
Using the strongest-performing creative identified in the A/B messaging test, we deployed paid ads with £20 over two days to test acquisition cost viability.

Ads were run across Meta platforms, tracking:

- Ad spend
- Click-through rate
- Waitlist sign-ups

Off/On	Campaign	Results	Cost per result	Budget	Amount spent	Impressions	Reach
	Hibearnate Campaign 1 Schedule Edit Duplicate	60 Link Clicks	£0.28 Per link click	Using ad set bu...	£17.02	2,389	1,998
	Instagram post: Be part of the dream to a bet...	5 Link Clicks	£0.22 Per link click	£20.00 Lifetime	£1.12	137	131
	Instagram post: Message us now.	— Link click	— Per link click	£20.00 Lifetime	£0.00	—	—
	Instagram post	3 Link Clicks	£0.61 Per link click	£20.00 Lifetime	£1.83	348	326
	Results from 4 campaigns	68 Link Clicks	£0.29 Per link click		£19.97 Total Spent	2,874 Total	2,421 Accounts Centre acco...

Cost per Waitlist Signup (CPWL) was calculated previously as: Total Ad Spend / Waitlist Sign-ups:

CAC							No. Conversions						CAC					
Spend	No. views	No. clicks	Cost /view	Cost /click	No. waitlists	Click/waitlist	85.00%	70.00%	55.00%	40.00%	25.00%	10.00%	85.00%	70.00%	55.00%	40.00%	25.00%	10.00%
1.21	198	4	£0.01	£0.30	1	0.25	0.85	0.7	0.55	0.4	0.25	0.1	£1.42	£1.73	£2.20	£3.03	£4.84	£12.10
20	3960	80	£0.01	£0.25	10	0.125	8.5	7	5.5	4	2.5	1	£2.35	£2.86	£3.64	£5.00	£8.00	£20.00

We then modelled estimated CAC under a 10% waitlist-to-paid conversion assumption. Estimated CAC remained below projected gross profit per unit of £20.3, indicating potential commercial viability under moderate conversion assumptions. Acquisition cost remains sensitive to real conversion rate and creative efficiency.

Next steps include:

- Extending campaign duration
- Testing additional creatives
- Measuring real conversion from waitlist to purchase

1.6 Risk Position Summary Table

Risk	Initial Risk Level	Current Position	Remaining Gap
Technical Feasibility	Very High	Medium	Scaling & durability
Installation Friction	High	Medium-Low	Larger demographic testing
Commercial Viability	High	Medium	Real purchase conversion

Across structured validation cycles, all three core risks have been reduced from high uncertainty to measurable, scenario-based viability:

- **Transparent-to-black PDLC** technology has been scientifically **fabricated** and **demonstrated** under controlled lab conditions.
- **Installation feasibility** has been refined through competitive benchmarking, iterative manual development, and real-user A/B testing.
- **Measurable sleep improvements** provide directional **quantitative evidence** of product value.
- **Acceptable price range** exceeds estimated cost base, **supporting sustainable gross margin.**
- Paid acquisition testing has generated measurable cost per waitlist signup and **modelled CAC** under multiple conversion scenarios.
- Organic content testing has identified **high-performing messaging** and demonstrated early top-of-funnel traction.

1.7 Evaluation

Despite meaningful risk reduction, several uncertainties remain:

- Conversion from waitlist to paying customer has not yet been observed.
- Small-batch production yield and consistency remain untested.
- UK manufacturing plant has high CAPEX.
- Long-term durability and environmental stress performance of PDLC require validation.
- (SOMETHING ABOUT INSTALLATION HERE!)
- Customer acquisition cost may fluctuate at scale and when targeting different demographics.

So, the immediate next-stage priorities are:

1. Launch a deposit-based pre-order mechanism to measure real purchase conversion (Kickstarter).
2. Conduct small-batch fabrication trials to evaluate production yield and repeatability.
3. Expand installation testing to uninstalling as well as a larger, more diverse user cohort.
4. Continue optimisation of paid creative to gain more traction, reduce CPWL and stabilise estimated CAC.

Financial Report

Executive Summary

This financial report presents Bear's high level financial plan and justifications for the next three years.

